

# Woochan (Andy) Jung

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## TECHNICAL SKILLS

**Languages:** JavaScript, Python, C/C++, TypeScript, HTML/CSS, Assembly

**Frameworks/Technology:** React, Node.js, NextJS, REST API, PyTorch, Flask, Tailwind CSS, SQLite, Rest API

**Developer Tools:** Git, Jenkins, Jira, Gerrit, VS Code, Visual Studio

**Libraries:** Pandas, BeautifulSoup4, OpenCV, Mediapipe, PyTorch, XGBoost, Numpy

## EDUCATION

**Simon Fraser University**

*Bachelor of Science in Computer Science*

Burnaby, BC

Sep 2023 – Present

## EXPERIENCE

**Software Engineer, Co-op**

January 2025 – Present

*OSI Maritime Systems*

*Burnaby, BC*

- Contributed to the development of **Vplus**, an internal error log visualization tool used by system engineers to diagnose and resolve bridge integration issues more effectively using **Python**.
- Independently designed and **Full-Stack** developed **SmartData.ai**, an AI-powered IDD error detection tool using **Flask** and **XGBoost**, reducing manual effort and cutting down and cost during bridge system development by **40%**.
- Enhanced core navigation software (**ECPINS**) by developing and testing new features to support situational awareness and operational efficiency for military vessels.
- Automated key workflows with **Python**, **PowerShell**, and **Jenkins**, improving reliability and reducing manual deployment errors.
- Collaborated with cross-functional Agile teams using **Jira**, and performed peer code reviews with **Gerrit** to ensure code quality and maintainability.

**Software Developer**

Sep 2024 – December 2024

*SFU Robot Soccer (Bandits FC)*

*Burnaby, BC*

- Designed field mapping and route optimization algorithms using **A\* pathfinding** for robot navigation.
- Enhanced software by developing new AI-driven agents using **C++** and **Python**.
- Virtually tested and optimized robot behaviour using graphical simulations using **QT**.

## PROJECTS

**ClearMyFridge | Python, Flask, PostgreSQL, C#, OpenAI API**

Jan 2025 – Present

- Fine-tuned two **GPT-based models** using curated datasets and prompt engineering for accurate recipe generation and ingredient rejection.
- Integrated the models via **Flask**-powered **REST API** for real-time usage through the mobile app using **Unity**.
- Implemented a **PostgreSQL** database to store user dietary restrictions and preferences, which were used to personalize recipe recommendations through continued textmodel fine-tuning as users interacted with the app.

**Grad Gains | NextJS, Typescript, React, Tailwind CSS, Git, NeonDB**

Jan 2024 – Present

- Built a finance web-app to help students invest through community-building features using **NextJS**.
- Implemented a **REST API** for real-time financial data and a dynamic dashboard.

**Syde-Kick AI | Python, OpenCV, PyTorch, Machine Learning**

May 2024 – Present

- Developed an MMA striking detection system with **Python** and **OpenCV**.
- Trained custom **CNNs** with **PyTorch** for enhanced strike detection accuracy.
- Built a responsive web interface using **React** and **TypeScript**, enabling users to analyze fight stats in real-time.

**Fit Pals | Computer Vision, Python, OpenCV, Mediapipe, Flask, SQL**

March 2024 – May 2024

- Developed a fitness tracking web application that provides users with feedback on exercise form and posture using **OpenCV** and **Mediapipe**.
- Built a push-up and squat repetition counter with real-time **Python**-powered feedback to improve user form.
- Implemented a web interface with **Flask** for users to track their progress and interact with feedback data.

## AWARDS

**1st Place, Hack the Sem:** Hackathon hosted by Google Developer Student Club.